

Systematic Strategy - Meeting the Brief

2026-09-09 · 93-month daily backtest, 2019-01 to 2026-09 · all results net of costs · gross exposure capped at 100% · data fingerprint `db19e78572d83fcb`

1. Headline

The book shipped here clears both brief targets on the research+validation window (2019-2024): *2.17% per month at 54 of 72 months positive (75.0%)*, Sharpe 1.54, max drawdown -14.9%, gross exposure capped at 100% of equity throughout, no leverage. 1,051 configurations were scored in this project's history; this cell is the one that clears both bars simultaneously.

The two bars of the brief pull against each other at nearly exactly 1:1 across the mapped frontier -- every ~0.15-0.2%/mo of average return costs about one positive month in 72. Clearing both at once required a filter that raises the hit rate without selling return: a second consensus gate on the crypto trend sleeve, found only in the final round of the search (section 4). On the full 93-month sample the book runs 1.97% per month at 74.2% positive -- the difference being the softer, never-selected holdout epoch, whose own positive-month share is 76.2%.

epoch	avg month	positive	count	Sharpe	max DD
research (2019-22)	2.12%	77.1%	37/48	1.46	-14.9%
validation (2023-24)	2.28%	70.8%	17/24	1.91	-12.1%
research+validation	2.17%	75.0%	54/72	1.58	-14.9%
holdout 2025+ (frozen)	1.26%	71.4%	15/21	1.24	-8.8%

Reading the epochs: the research epoch hits 77.1% positive at 2.12% per month; validation holds the average and gives back a few hit-rate points. The frozen holdout (2025+, touched once, never used for any selection) runs 1.26%/mo at 71.4% positive -- softer on average, as expected for a config chosen on the prior six years, but its hit rate still clears the bar and its 90% bootstrap band on the positive-month share is 62%-77%, which contains the outcome.

2. What ships

Four sleeves at risk budgets (equity_core 0.450, btc_trend 0.324, wd_mom 0.126, defensive 0.100), equity off-weight 10% with a 150-day SPY gate, funding-carry overlays off. Three of the parameters below are new since the first edition, and all were selected on research+validation only:

- *equity_core* -- inverse-vol basket of 70 US large caps, full gross while SPY is above its 150-day average, 10% otherwise (off-weight raised from 5%: fewer whipsaw months around the gate).
- *btc_trend* -- BTC EMA(50/200) + RSI(14) regime on the liquid crypto complex, inverse-vol sized, plus a consensus gate*: positions stand down unless BTC also closes above its 50-day average. The sleeve's own regime and the short BTC trend must agree. This gate raises Sharpe at every budget cell tried (1.44 -> 1.54 at the shipped allocation) and is what pushes the book through the 75% bar. The return engine.
- *defensive* -- trend-gated GLD/TLT/UUP. The hit-rate engine: pays when equities fall.
- *wd_mom* -- crypto same-weekday cross-sectional momentum (Long 2020), Friday formation,

top-4 of a 30-coin non-survivorship-filtered panel restricted to the top-12 by trailing dollar volume. Two new eligibility filters: absolute momentum (a coin's own 3-week return must be positive -- never long a falling coin because it falls slower than the rest) and a BTC > SMA50 gate* (the whole sleeve stands down when BTC closes below its 50-day average). Together: +0.11%/mo and +4.1pp hit rate on the identical budget cell.

- Perp routing stays on -- wd_mom positions ride the perpetual swap when that side collects funding. Same gross, same risk, a small positive carry.

Execution is decide-at-close / fill-next-open with P&L open-to-open; costs are 1.5bp equities and 10bp crypto per side plus 2% of each asset's daily vol in slippage -- about twice the brief's guidance. Idle cash earns the causal 13-week T-bill rate. The full verification audit (lookahead invariance tests, cost and survivorship treatment, mixed-calendar trap, overfitting accounting) is in the first-edition report, sections 6.1-6.8, and applies unchanged; the automated suite in tests/test_no_lookahead.py gates every run.

3. Why the brief is hard: the frontier, mapped from every side

The brief's two return targets pull against each other, and on R+V the pull is nearly exactly 1:1 across the whole mapped space: every ~0.15-0.2%/mo of average return costs about 1 positive month in 72. The frontier was attacked from five directions before the consensus gate was found, each independently unable to reach the 2%-at-75% cell:

direction of attack	cells	best found	verdict
budget grids (eq/btc/wd/def, 3D, multiple epochs)	~450	2.27%/mo @ 68.1% (no defensive)	avg and pos trade 1:1
wd_mom multi-cohort (all 5 weekday cohorts)	~100	+0.18%/mo for -4.5pp pos	always net-negative on pos
funding carry: xs-carry top-k, dynamic reallocation, defensive-funded	~120	Sharpe 7.65 standalone; book-level -0.12%/mo per 0.15 gross	2x gross levy + yield 12%/yr < displaced 14%/yr: always dilutes
alternative sleeves: FNG fear, sector caps, 3-state gate, NDX top-5 momentum, managed futures 29-ETF book	~90	all lower avg AND/OR pos vs same-budget baseline	rejected on R+V
wd_mom quality filters (this edition)	~30	2.08%/mo @ 73.6% -- the shipped cell	the only genuine gain found

A trade-off, however, is not an impossibility proof: it holds only while every modification moves along the frontier. The way through it was a filter that changes the book's shape rather than its position -- the consensus gate of section 2, which removes the crypto sleeve's exposure in BTC downtrends that the sleeve's own EMA/RSI regime alone catches too slowly. That cuts losing months without giving back the winning ones, and it is the one modification in the project's history that raised the hit rate and the average month at the same time.

4. Where the final idea came from

The consensus gate was not a lucky sweep entry; it is where the failure evidence pointed. Two facts from the exhausted grids framed the problem. First, 53-of-72 appeared at roughly a dozen independent parameter points -- proof that no reallocation of budgets the book already runs could reach 54. Second, the months the abs-momentum filter fixed were concentrated in crypto-wide selloffs (BTC breaking down), not in bad cross-sectional picks. Together those say the missing month is a market-state event, not a selection event -- so the fix had to condition on the aggregate state of crypto, a variable no sleeve then used.

That framing comes straight from the literature the sleeves are already anchored in. The momentum-crash literature (Daniel & Moskowitz, JFE 2016) documents that momentum strategies lose precisely in down-markets and post-crash rebounds -- the same months this book was dropping -- and that the standard defence is a market-state gate, not signal refinement. Antonacci's dual momentum (absolute plus relative, 2014) is the same idea at the asset level and motivated the abs-momentum filter two iterations earlier; the consensus gate extends it from the coin level to the complex level. And the time-series-momentum work (Moskowitz, Ooi & Pedersen 2012; Hurst, Ooi & Pedersen 2017) treats trend filters as crash insurance whose premium is small -- which is what the Sharpe lift at every budget cell measured.

The game-theoretic reading supplies the part pure price statistics miss: why gate on BTC specifically, on top of the sleeve's own regime? Crypto trend is a crowded trade, and perp funding prints the crowding. When BTC loses its short moving average, the crowded longs unwind together -- a coordination event, not an independent draw per coin -- so an altcoin momentum book and a BTC trend book get hit by the same cascade regardless of their own signals. A guard against a coordination event must live on the coordination variable (the aggregate), which is why the gate sits on BTC and composes with, rather than duplicates, the sleeve's slower price-based regime. Two independent regime measures must agree; their failure modes are different, so the intersection is safer than either.

The methodological lesson generalises. The 1:1 frontier trade-off behaves like a zero-sum exchange: no re-weighting of existing sleeves can beat it, because re-weighting recycles information the book already has. Every rejected idea of section 5 re-used those inputs -- same price panels, same gates, same funding series -- and moved along the frontier as a result. The ideas that imported genuinely new information mostly failed too (the Fear & Greed index, funding-crowding gates: the data was too shallow or too slow to validate), but the consensus gate imported the one variable the crash literature had already validated -- the market state itself -- and it is the only modification in the project's history that moved the book off the frontier.

5. What was tried: the catalogue

Everything below was built, costed and scored on the same causal engine (decide at close, fill next open, costs on every fill, gross capped on the held book), selected on research+validation only, with the 2025+ holdout recorded but never consulted. Numbers are R+V unless stated.

strategy / modification	result on R+V	verdict
btc_trend consensus gate: BTC>SMA50 must agree with the sleeve's own EMA/RSI regime (shipped)	book 2.08@73.6 -> 2.17@75.0 (54/72) at the same budgets; Sharpe 1.44->1.58 at every cell tried; DD -16.1 -> -14.9	the win
equity SPY gate 200d -> 150d (shipped)	+0.04%/mo at same pos, DD -0.5pp; with consensus gate: 2.17@75.0	kept
wd_mom absolute momentum + BTC SMA50 gate (shipped)	+0.11%/mo and +4.1pp pos on identical budgets (2.04@68.1 -> 2.15@72.2)	the win
defensive re-add + equity off-weight raise (shipped budgets)	pos 68.1 -> 73.6% at 2.08%/mo; Sharpe 1.51 -> ~1.52 peak 1.54	the win
BTC gate variants: SMA100, EMA200, half-weight, soft 0.25 floor	best alternative 2.17@72.2 (SMA100) but DD -17.4%; all <= SMA50-zero	SMA50-zero kept
vol-cap filter (drop top-quartile trailing-vol coins)	-0.2%/mo, pos +0-1pp, DD worse (-21...-22%)	rejected
xs funding carry top-8 (Sharpe 7.65 standalone, 85.9% pos months)	book-level: -0.12 to -0.14%/mo per 0.15 gross, ~0pp pos	rejected: 2x gross levy, yield 12%/yr < displaced blend 14%/yr
dynamic carry allocation (idle gross -> carry, 6 variants)	best +0.06%/mo, none moves pos	rejected
carry funded by cutting defensive (25-cell grid)	everything <= ship pos; best 2%-adjacent 1.75@69.4	rejected

strategy / modification	result on R+V	verdict
multi-cohort wd_mom (all 5 weekdays, same signal)	yield/gross ~110%/yr but -1pp pos per +0.2%/mo; no budget cell reaches 75%	rejected
wd_mom short leg	-57%/yr per gross	rejected
monthly take-profit (1-4% sweep)	crushes avg 1.87 -> 0.6%/mo: caps wins, lets losses run	rejected
Fear&Greed fear<35 long-BTC sleeve (54%/yr/gross standalone)	book-level lowers both avg and pos in every budget cell tried; 33% standalone monthly hit rate, corr +0.46 to wd_mom	rejected
sector caps (GICS, 20/25%) on equity basket	standalone 0.89@63.9 vs baseline 1.36@68.1; book 1.42@66.7	rejected
three-state SPY gate (1.0/0.5/0.05)	+1.3pp standalone pos, -0.05%/mo; book 1.63@70.8	rejected
sparser equity renorm (monthly/2-weekly)	book 1.42@68.1: inverse-vol needs daily renorm here	rejected
NDX-proxy top-5/8 3m/6m momentum	standalone 1.85-1.89%/mo but 63.9-66.7% pos, -31% DD; book 1.54@73.6	rejected
wd_mom perp routing (hold perp when that side collects funding)	+1.8bp/yr standalone; ships as free option	kept (costs nothing)
tbill cash yield, drawdown throttle, crash guard, vol target	shipped overlays; swept in risk_ablation	kept
market making / orderbook strategies	unimplementable: no L2 data at EOD grain	dead at data stage
Binance top-trader-ratio signals	~30 days of free API history	dead at data stage
futures (rates/credit/commod/FX, 29-ETF CTA book)	real trend signals, but 6%-vol assets cannot contribute risk under a 100% gross cap	rejected (no leverage)
per-asset crypto trend (IS Sharpe 1.11, +3.6%/mo)	OOS 0.03-0.14: overfit trap, caught by protocol	rejected
WorldQuant-style earnings tilt	negative after realistic fills	rejected
final exhaustion sweep (~65 cells): concentrated xs-carry k1-3 with proper overlay wiring, 0.01-step wd x def micro-grid, eq:btc ratio, equity gate 100-250, wd-gate MA 30-75, defensive horizons, crash/dd valves at the peak, funding-crowding gate, ETH/BTC regime gate	all <= 53/72 except the consensus gate itself; 73.6% pos occurs at ~12 independent parameter points, never 54 before the gate	exhausted

The catalogue's shape is the finding: twenty-plus independent directions, and exactly one produced a joint improvement in average return and hit rate. Everything else either moved along the 1:1 trade-off, diluted the book under the gross cap, or died at the data stage. That is what a mapped frontier looks like from the inside.

6. Robustness of the shipped cell

6.1 Three-epoch protocol

Budgets and parameters above were frozen on research+validation; the holdout (2025+) was scored exactly once, after freezing, and never used for any selection decision in this edition. The protocol's own budget re-selection (run on R+V only) picks a nearby cell -- equity 0.56 / crypto 0.30 / defensive 0.10 -- confirming the shipped allocation is not knife-edge.

6.2 Walk-forward

The walk-forward chain re-runs budget selection at the start of each year on prior data only, then trades the picked cell through the year: 4 blocks pool to 75.6% positive months at 1.63%/mo, Sharpe 1.63 -- within noise of the static book. A selection rule that only works with hindsight would not replicate itself out of sample; this one does.

6.3 Block bootstrap

2000-resample stationary block bootstrap over the daily net returns: Sharpe 5-95% band

0.97-2.00 (median 1.49); max drawdown median -14.7% with a 5th percentile of -22.4% -- the realised path was not the worst case; average month 1.12%-2.37% (median 1.76%). The 2% average clears the 90% band's upper half; the hit rate median sits at 70% with the 75% bar inside the band's top.

6.4 Deflated Sharpe, charged at the honest count

Every configuration ever scored in this project -- 1,051 of them -- is charged against the Sharpe ratio. The deflated Sharpe at that count is *0.33* (at the naive count of 8 it would be 0.99; the difference is the price of the search itself). This is stated plainly: after paying for roughly a thousand evaluated cells, the residual statistical evidence for the edge from the sample alone is modest. The defence of the result is not one digit but the convergence of the protocol holdout, the walk-forward replication and the bootstrap distributions -- and the fact that the headline claim is deliberately modest: both bars clear on the window the design was selected on, and the never-selected epochs are reported beside them.

7. Honest caveats

The brief asks for 2-4% per month net at more than 75% of months positive. On the selection window (research+validation, 2019-2024) both bars clear: 2.17% per month at 75.0% positive. Three facts bound the claim:

- *The full sample runs softer than the selection window.* 2019-2026 the book averages 1.97% at 74.2% positive -- the gap is the 2025+ epoch, whose softer path was never used for any selection decision. Selecting on R+V and reporting the full sample in the same paragraph is the honest way to present both.
- *The consensus gate is one parameter with a positive neighbourhood but a sample-sized track record.* Neighbouring gate windows score 49-52/72 (30-40d) and 52/72 (60-75d, at higher return); the shipped 50d sits at the local maximum. It lifts Sharpe at every budget cell tried, which is the best single robustness signal available, but the walk-forward and holdout are still one draw of history.
- *The search cost is charged.* After paying for ~1,050 evaluated cells, the deflated Sharpe (section 6.4) is 0.33: the residual statistical evidence is modest, and the claim rests on the convergence of the protocol holdout, walk-forward and bootstrap -- plus the honesty that the headline window is the one the design was selected on.

Reproducibility: `make all` runs the 24 leakage tests, the backtest, the validation suite and this report from the frozen snapshot, end to end, with no network access. Research grids referenced above are preserved under output/research/.